



LEARNING OBJECTIVES

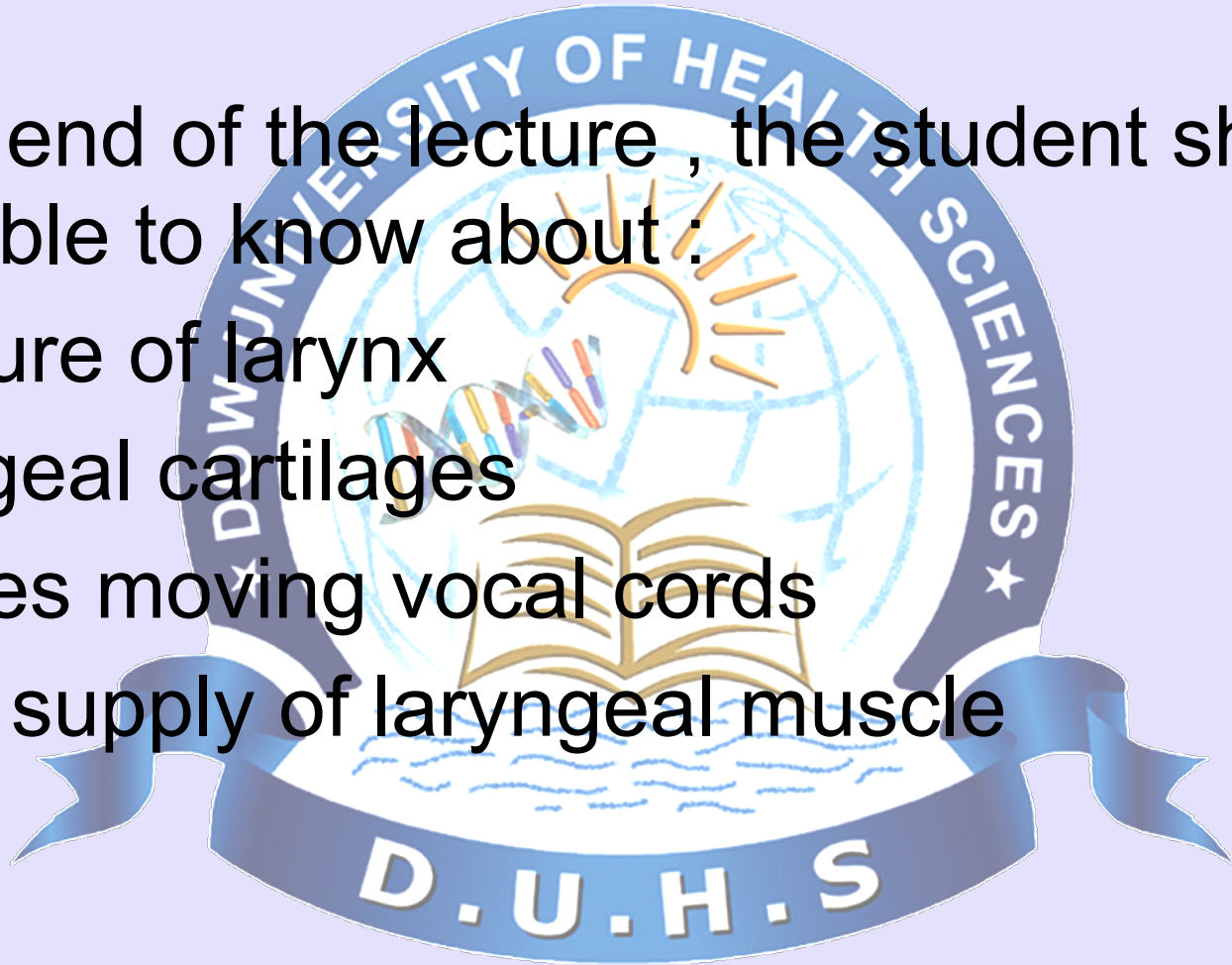
At the end of the lecture , the student should be able to know about :

Structure of larynx

Laryngeal cartilages

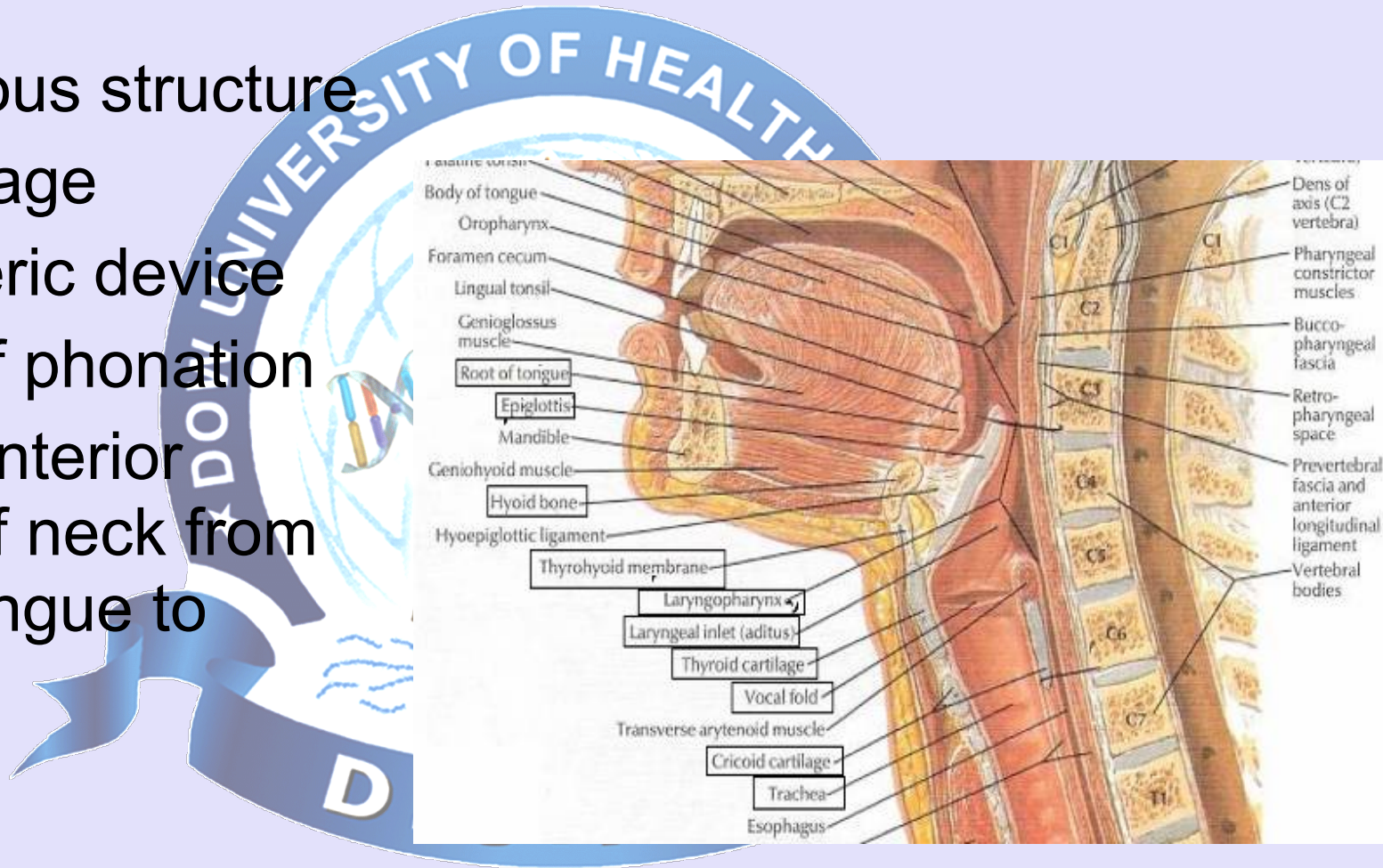
Muscles moving vocal cords

Nerve supply of laryngeal muscle



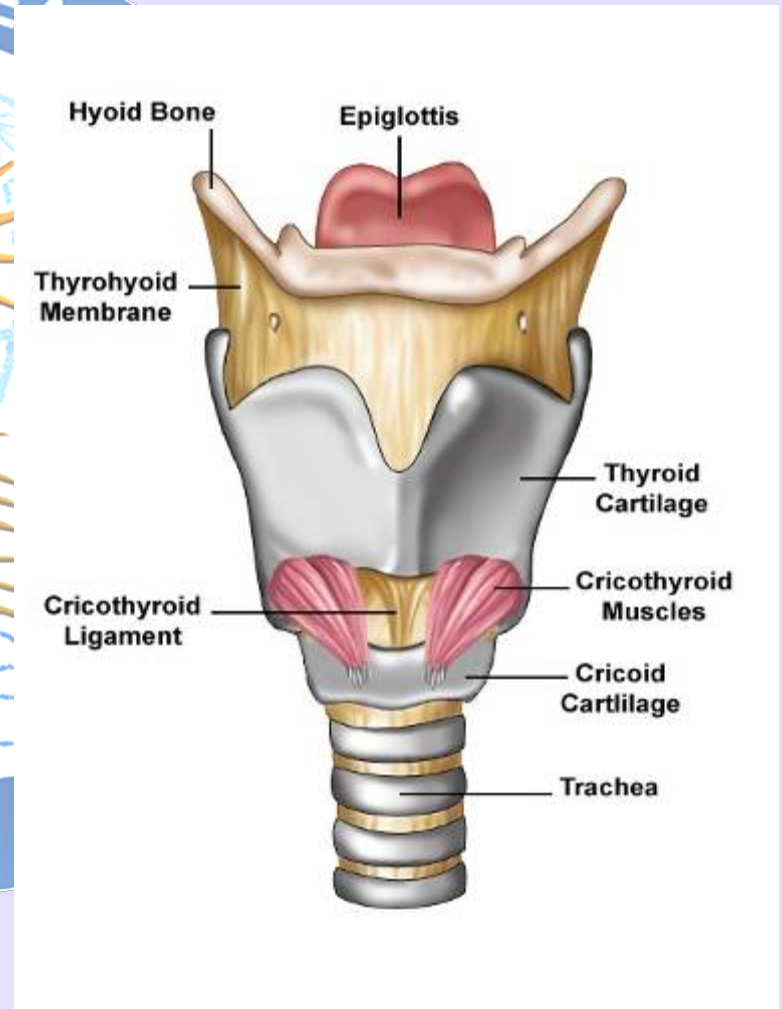
LARYNX

- Musculo ligamentous structure
- Air passage
- Sphincteric device
- Organ of phonation
- Lies in anterior midline of neck from root of tongue to trachea



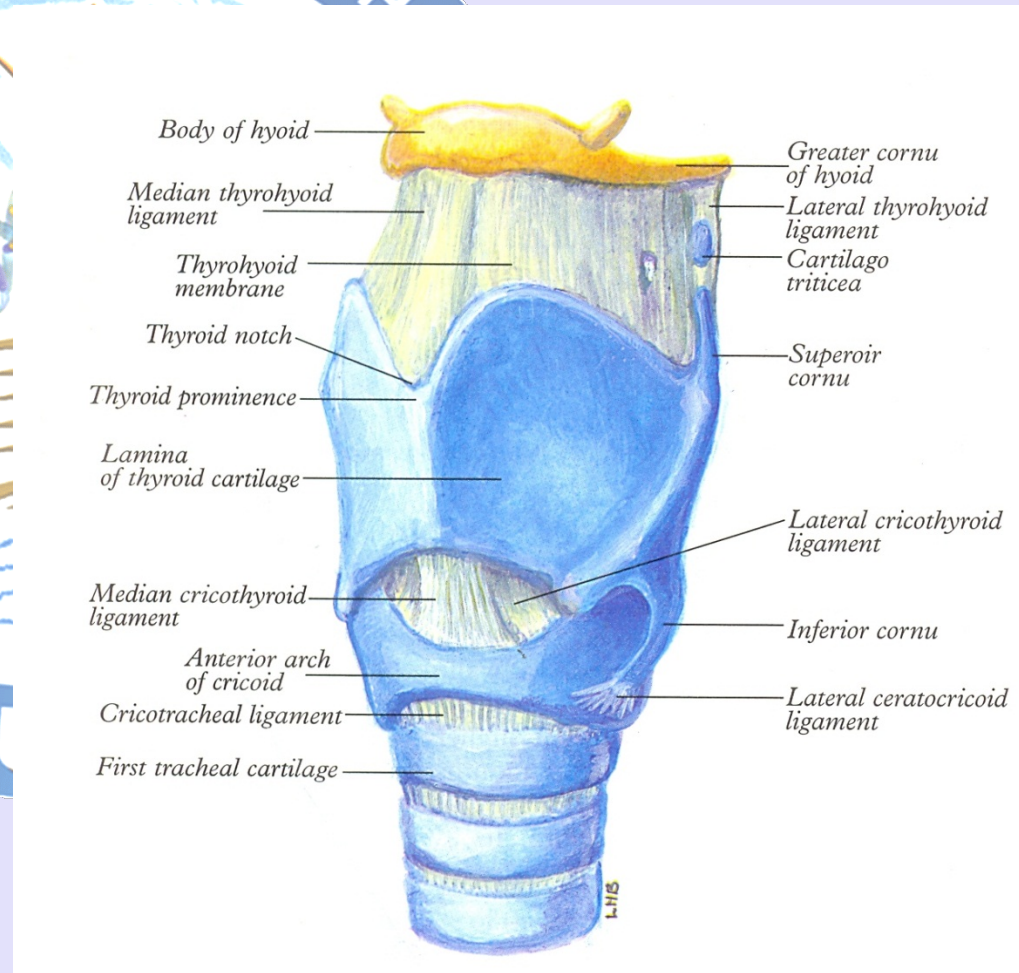
LARYNX

- Open above in laryngopharynx
- Continuous below in trachea
- Lies at the level of C 4, C5, C6 vertebrae
- Covered in front by skin, fascia and infrahyoid muscles
- Covered on sides by thyroid gland
- **Length** 44mm(m), 36mm(f),
- **TS** – 43mm(m), 41mm(f),
- **Sagital** diameter 36m and 26F
- Male larynx enlarges after puberty



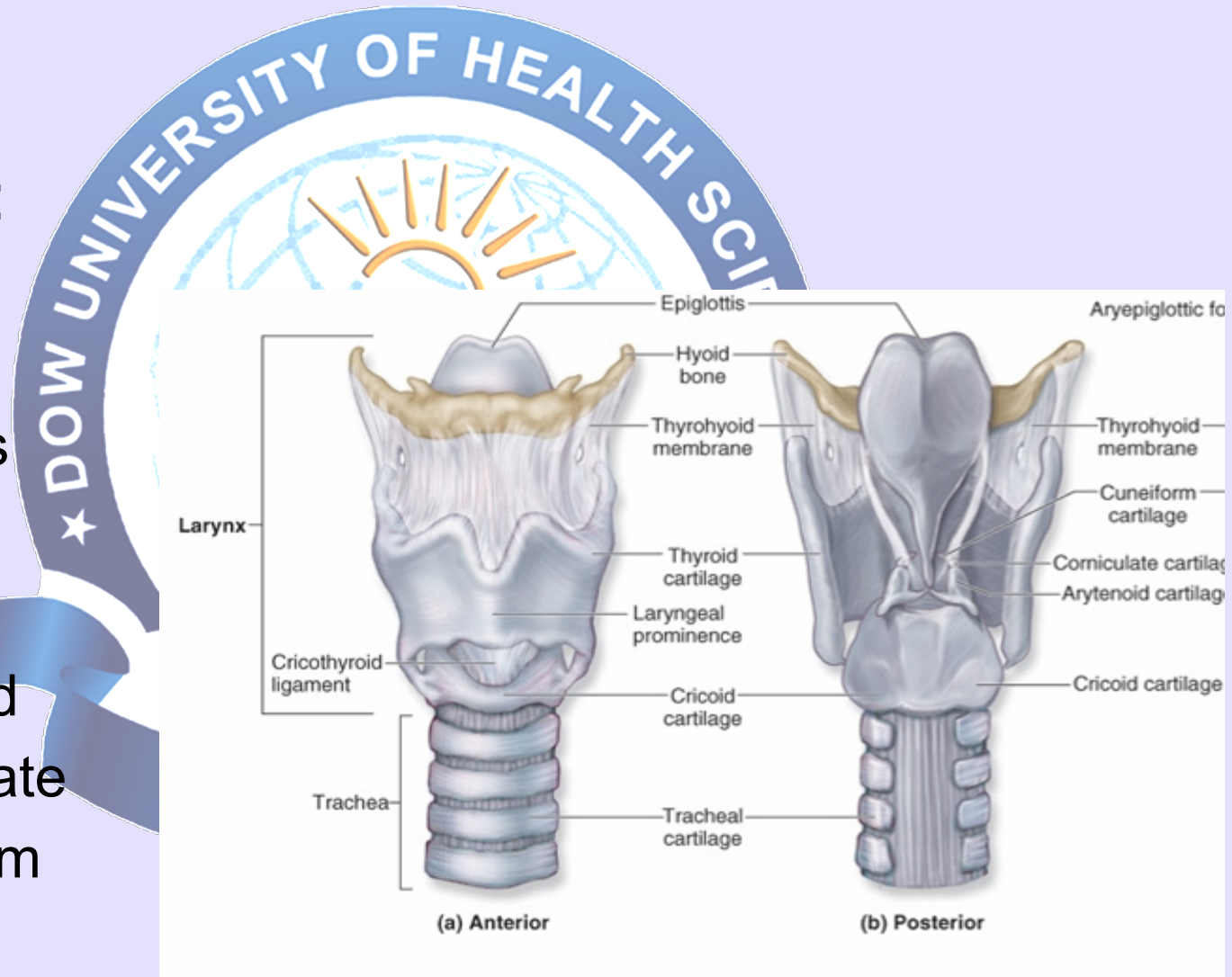
STRUCTURE OF LARYNX

- Cartilagenous framework
 - Paired
 - Unpaired
- ligaments and membranes
 - Extrinsic
 - Intrinsic
- Muscles
 - Extrinsic
 - Intrinsic
- Mucosa
 - Inlet of larynx
 - Cavity of larynx



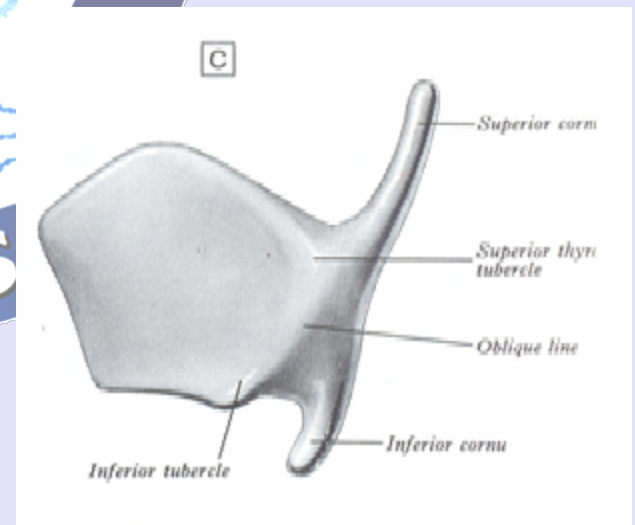
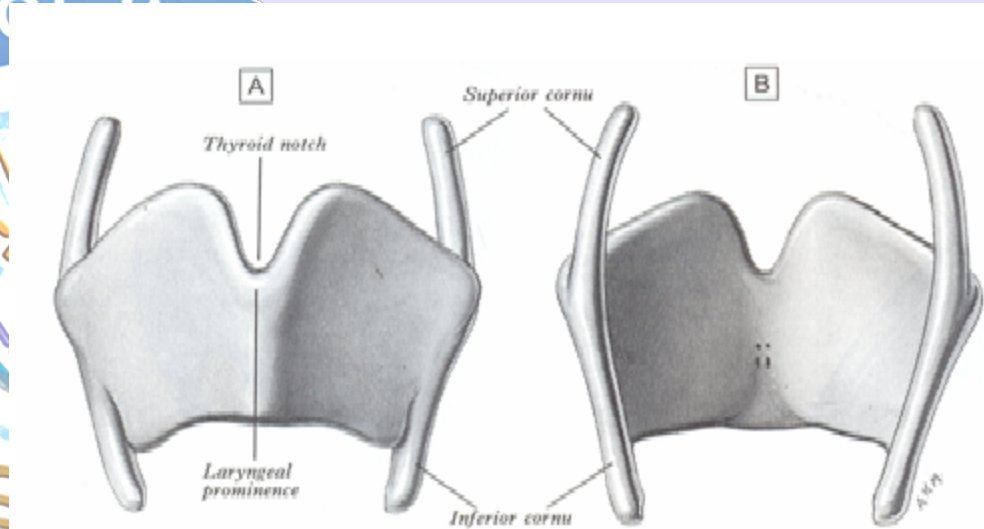
CARTILAGES OF LARYNX

- Unpaired cartilages:
 - Thyroid
 - Cricoid
 - Epiglottis
- Paired cartilages
 - Arytenoid
 - Corniculate
 - Cuneiform



THYROID CARTILAGE

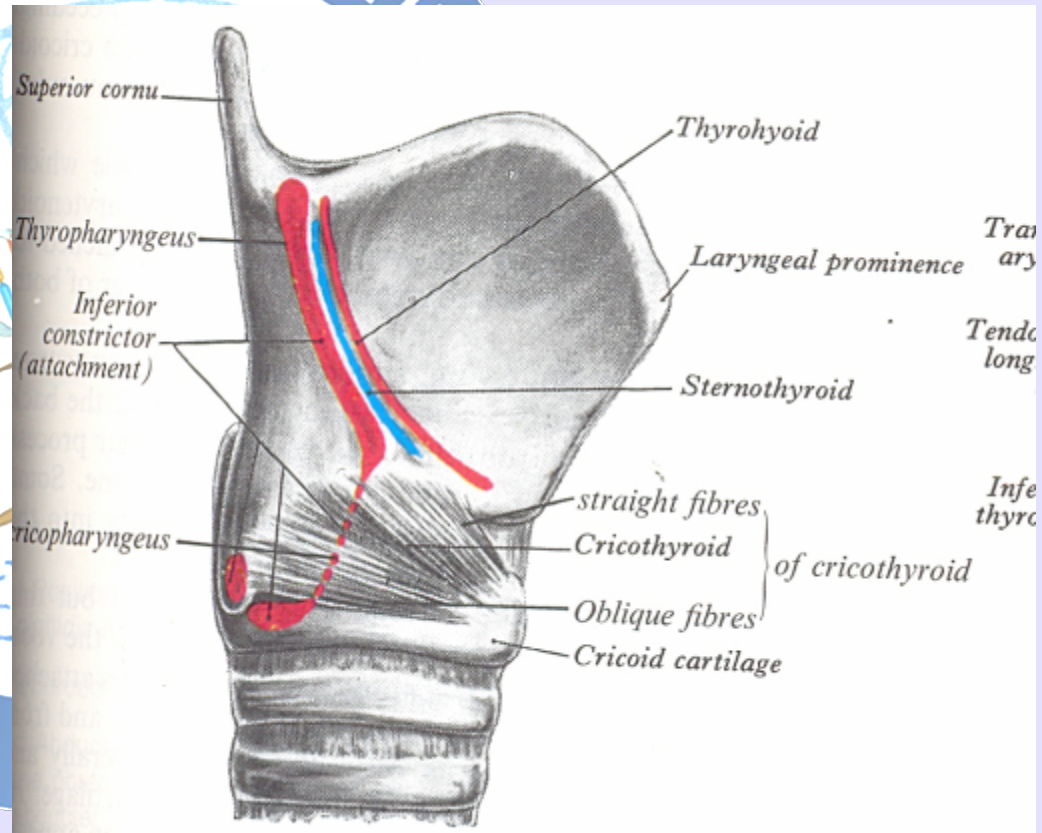
- Largest laryngeal cartilage
- Hyaline
- Two lamina
- Join at midline at V angle
- Thyroid notch between anterior borders of lamina
- Posterior borders extend upward into superior cornu and downwards into inferior cornu
- Anterior surface possess an oblique line



THYROID CARTILAGE

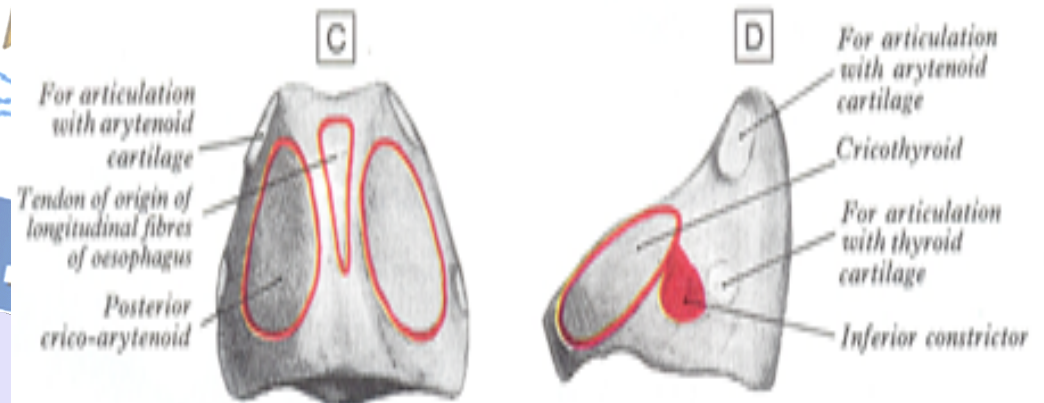
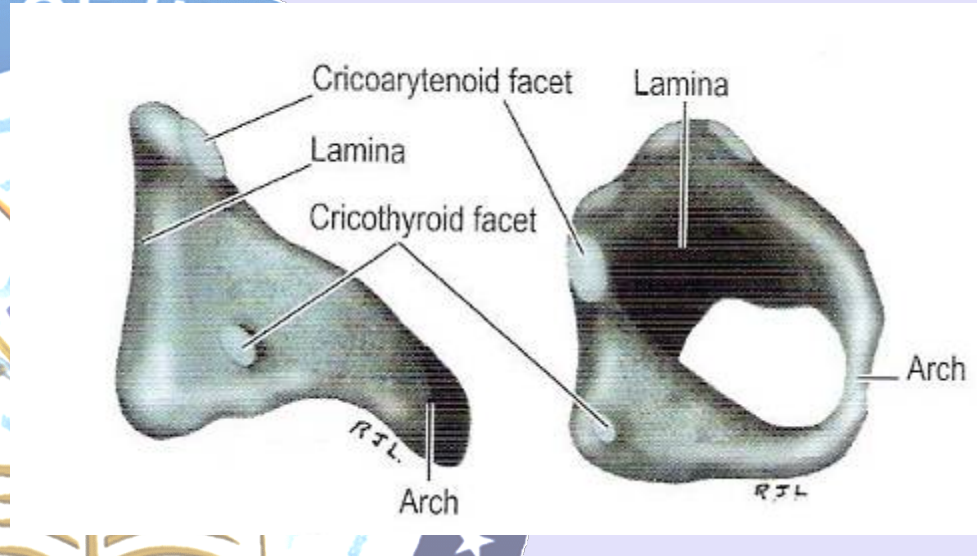
Attachments

- Superior cornu attached with greater horn of hyoid bone
- Inferior cornu articulates with cricoid at cricothyroid joint
- Oblique line provides attachment to
 - inferior constrictor muscle of pharynx
 - Thyrohyoid muscle



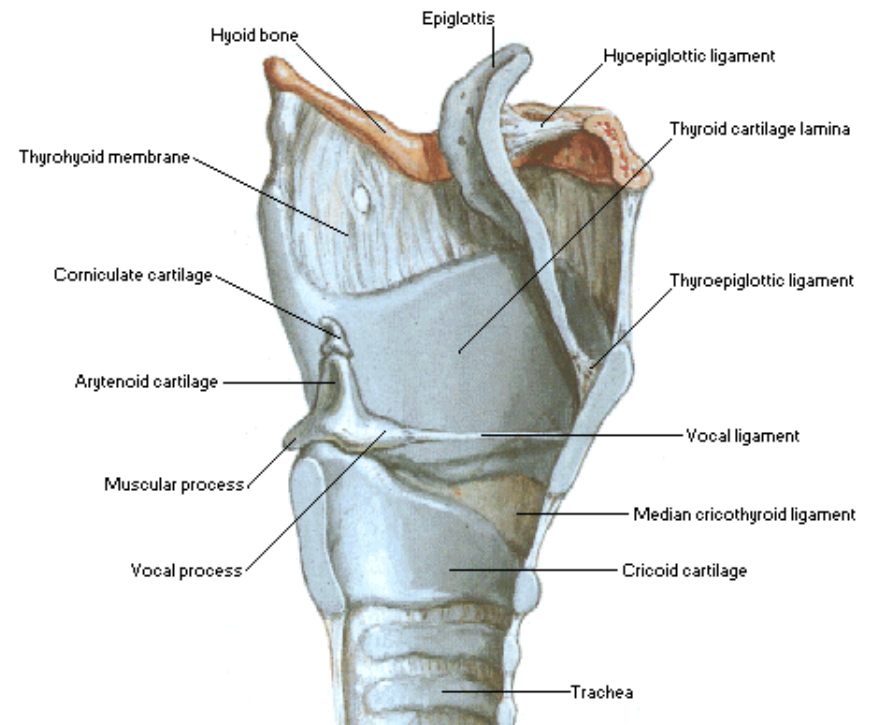
CRICOID CARTILAGE

- Ring shaped
- Hyaline cartilage
- Lies below thyroid cartilage
- Broad plate behind, shallow arch in front
- On lateral surface facet for articulation with thyroid
- Posteriorly, on upper border facet for arytenoid



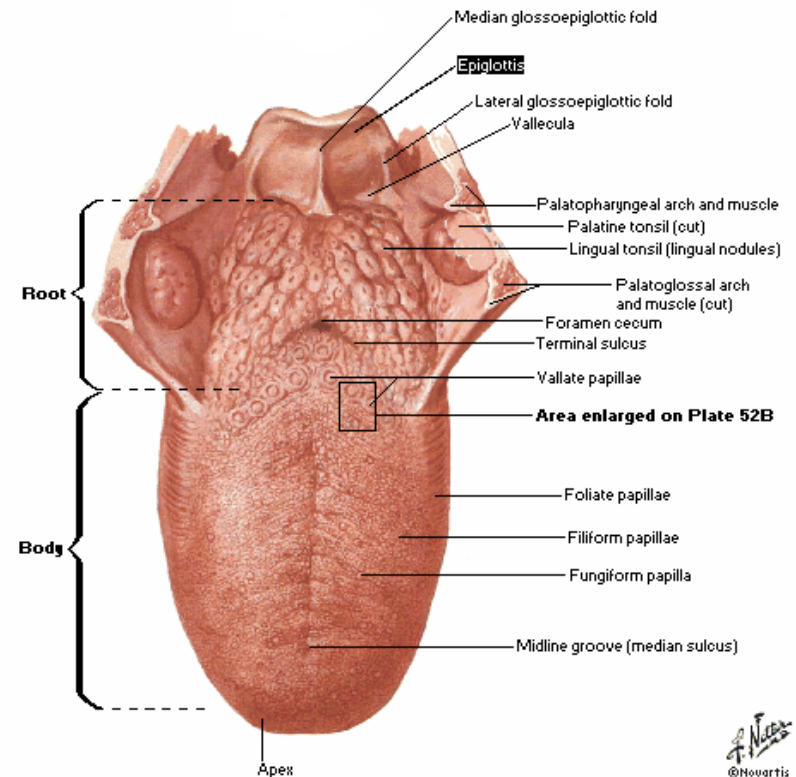
EPIGLOTTIS

- Leaf shaped
- Elastic cartilage
- Lies behind the root of tongue; in anterior wall of upper pharynx
- Upper end broad and free
- Lower end (stalk) pointed; attached on the posterior surface of angle of thyroid cartilage
- Has anterior and posterior surface



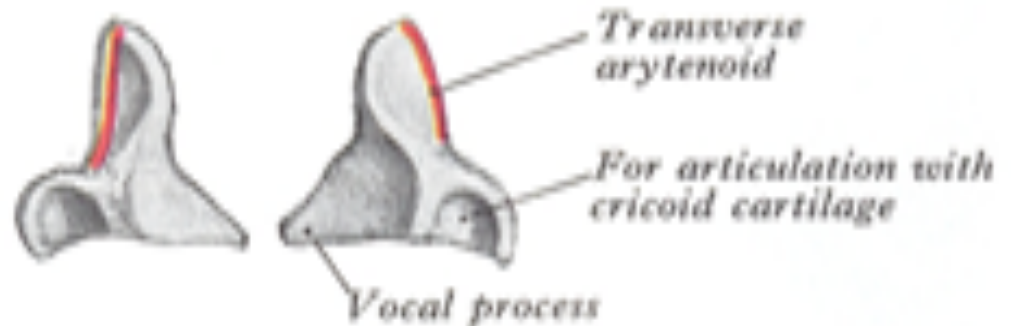
EPIGLOTTIS

- Attached to arytenoid cartilages by aryepiglottic fold
- Covered on both surface by mucosa
- Mucosa passes forward onto tongue as median glossoepiglottic fold
- On each side is vallecula
- Laterally is lateral glossoepiglottic fold

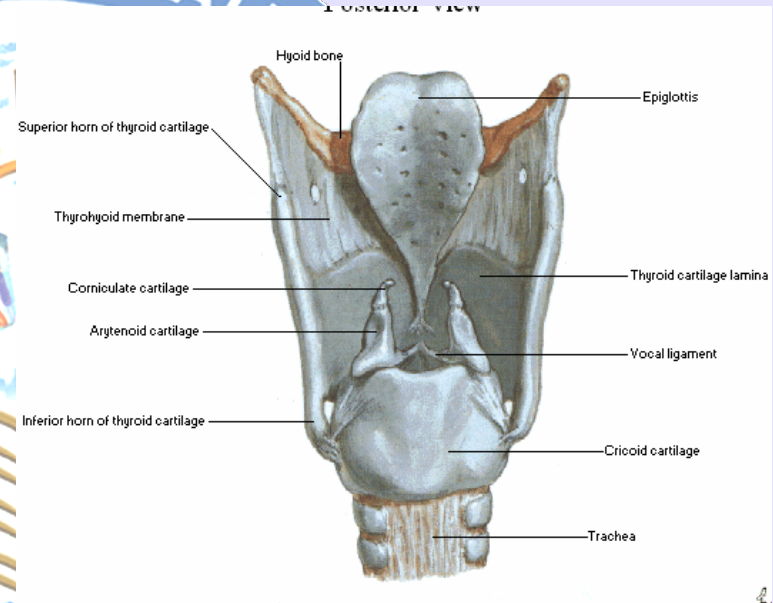


ARYTENOID CARTILAGES

- Paired
- Pyramid shaped
- At the back of larynx
- Only tip is elastic; rest of the body is hyaline
- Possesses :
 - Apex
 - Base
 - Vocal process
 - Muscular process
 - Anterior
 - Medial
 - Posterior surface



- Apex articulates with small corniculate cartilage
- Base articulates with upper border of posterior lamina of cricoid
- Vocal process give attachment to vocal ligament
- Muscular process give attachment to posterior and lateral cricoarytenoid muscle

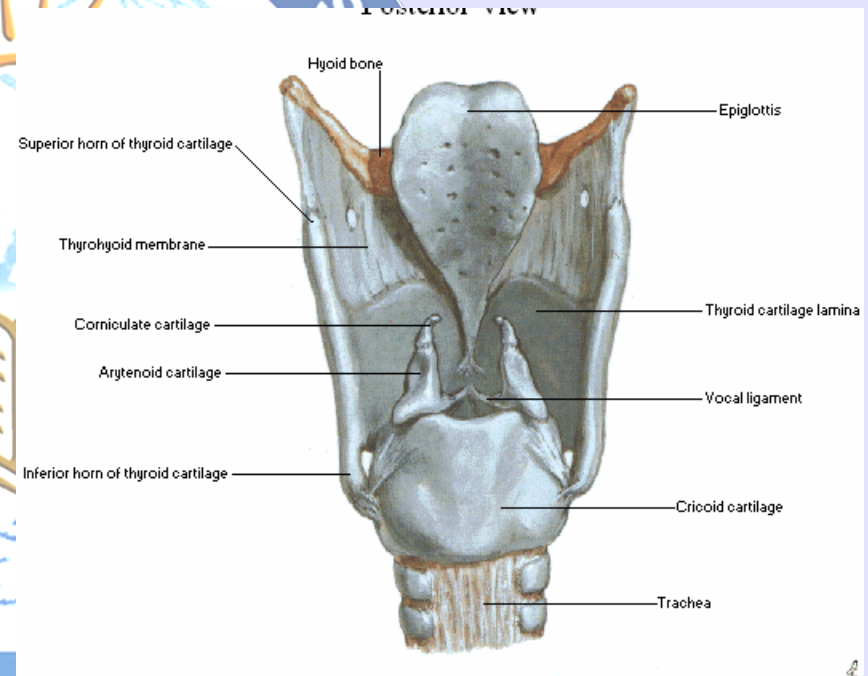


Corniculate:

- Conical nodules
- articulate with the apex of arytenoid cartilage
- Attachment to aryepiglottic folds

Cuneiform:

- Rod shaped pieces placed in aryepiglottic folds



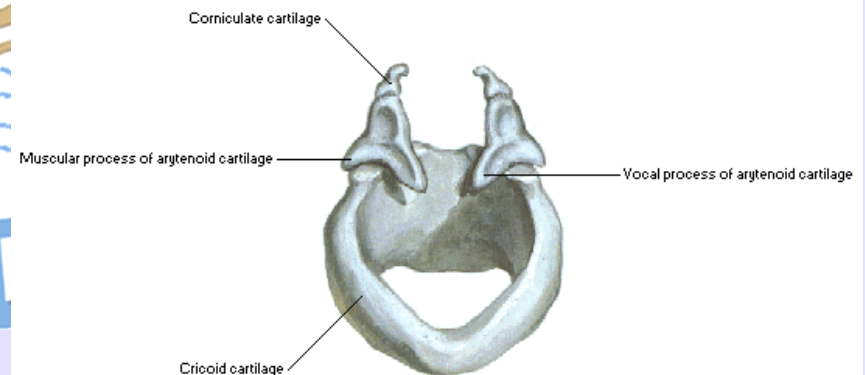
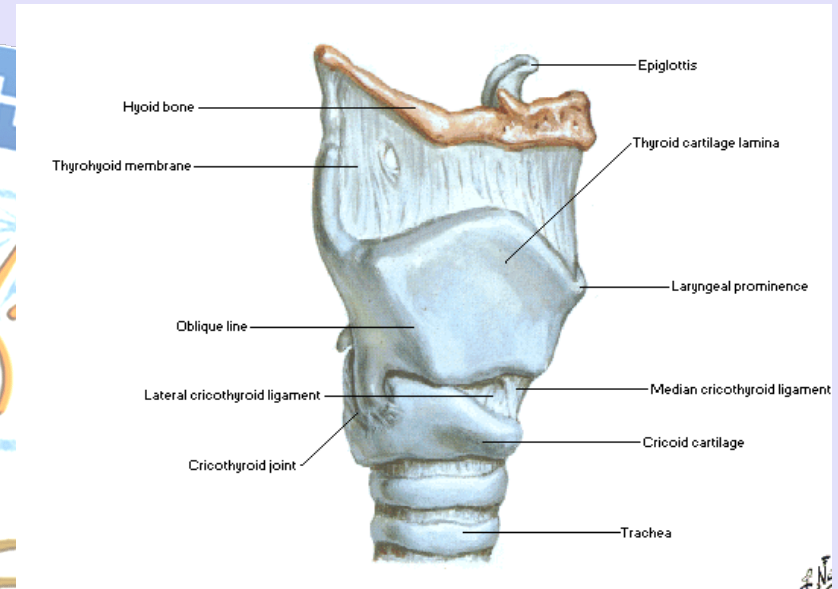
JOINTS OF LARYNX

Cricothyroid:

- Synovial
- Between inferior cornu of thyroid cartilage and side of cricoid cartilage
- Rotatory and gliding movements in transverse axis.

Cricoarytenoid:

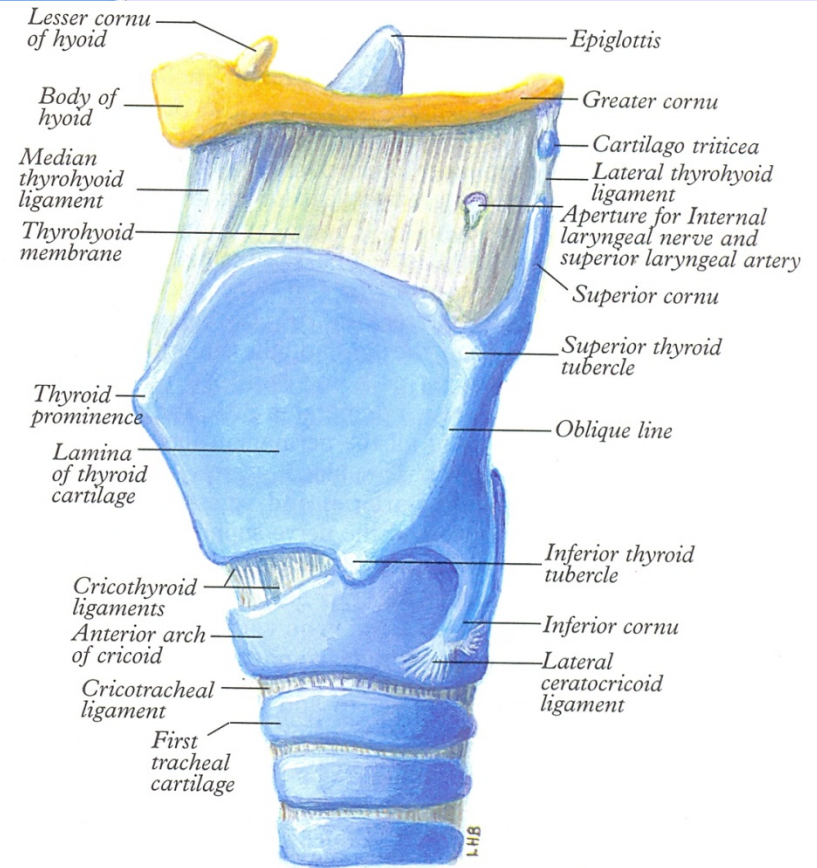
- Synovial
- Between base of arytenoid cartilage and the upper border of lamina of the cricoid cartilage
- Rotatory and gliding movements in vertical axis



MEMBRANES AND LIGAMENTS OF LARYNX

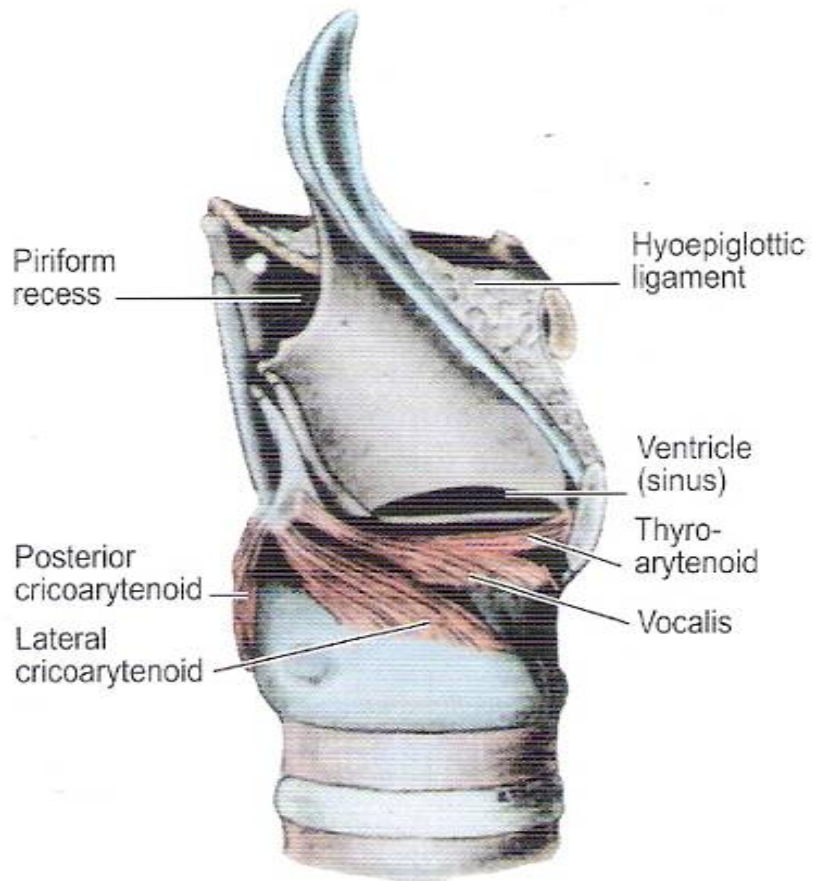
Thyrohyoid membrane:

- between thyroid cartilage and hyoid bone
- median and lateral part form **median and lateral thyrohyoid ligament**.
- pierced by internal laryngeal nerve and superior laryngeal vessels.



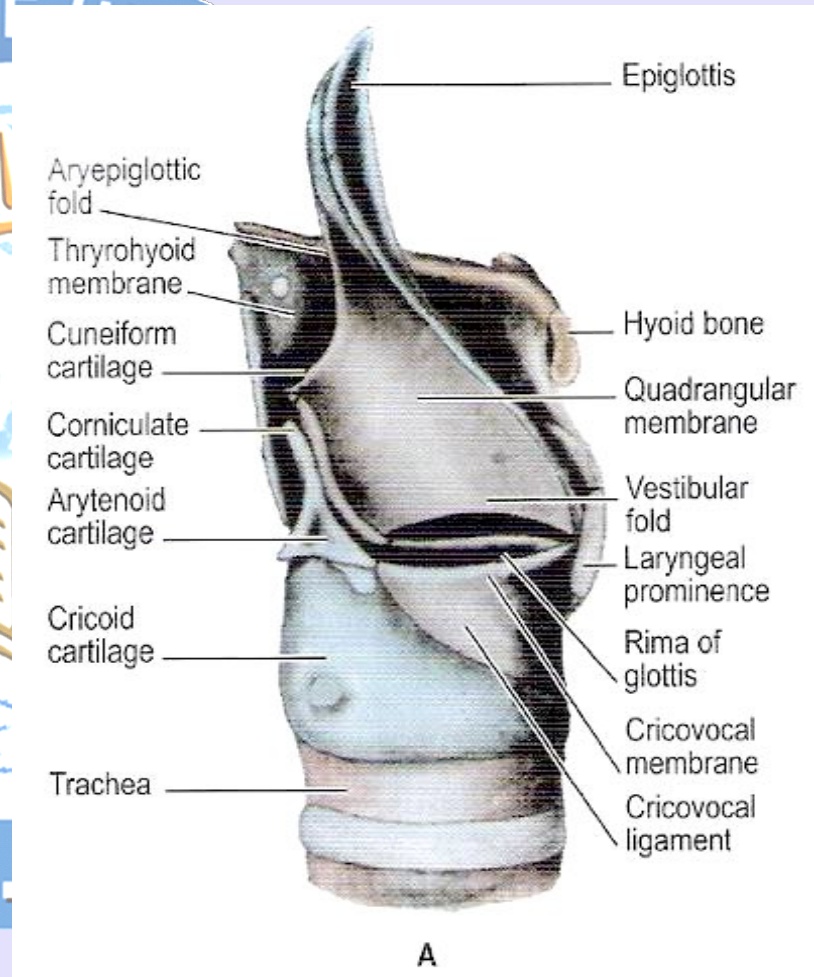
MEMBRANES AND LIGAMENTS OF LARYNX

- **Hyoepiglottic lig.:**
Between epiglottic cartilage and hyoid bone.
- **Cricotracheal lig.:**
Cricoid cartilage and trachea(1ST RING).



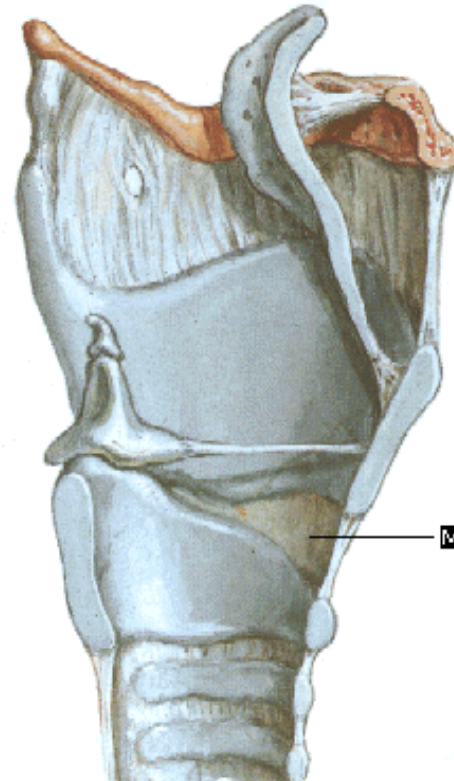
MEMBRANES AND LIGAMENTS OF LARYNX

- QUADRANGULAR MEMBRANE
- Extends between epiglottis and arytenoid cartilages
- Thickened inferior margin forms vestibular ligament



MEMBRANES AND LIGAMENTS OF LARYNX

- Cricothyroid ligament
- Lower border at cricoid
upper border
- Upper margin, elastic,
forms vocal ligament
- VOCAL LIGAMENT
- Anterior end attached to
thyroid cartilage
- Posterior end attached
to vocal process
arytenoid cartilage



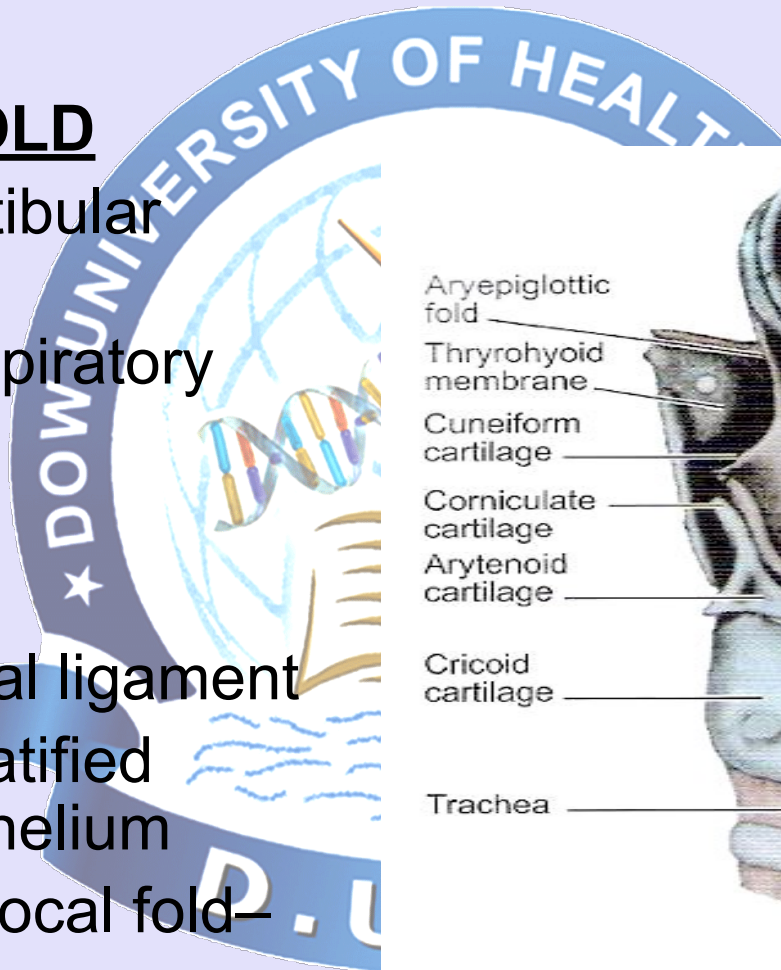
LARYNGEAL FOLD

VESTIBULAR FOLD

- formed by vestibular ligament
- Covered by respiratory epithelium

VOCAL FOLD

- Mobile
- Formed by vocal ligament
- Covered by stratified squamous epithelium
- Gap between vocal fold—rima glottis



A

INLET OF LARYNX

Backwards and upwards

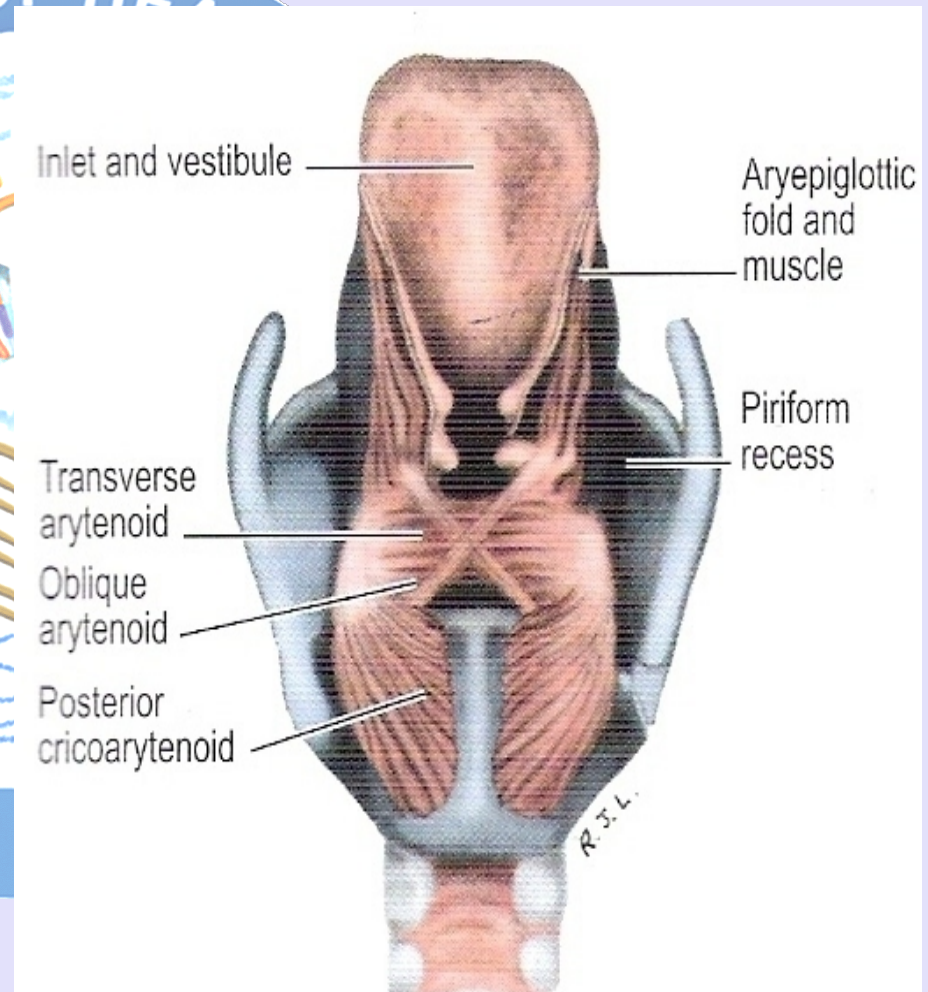
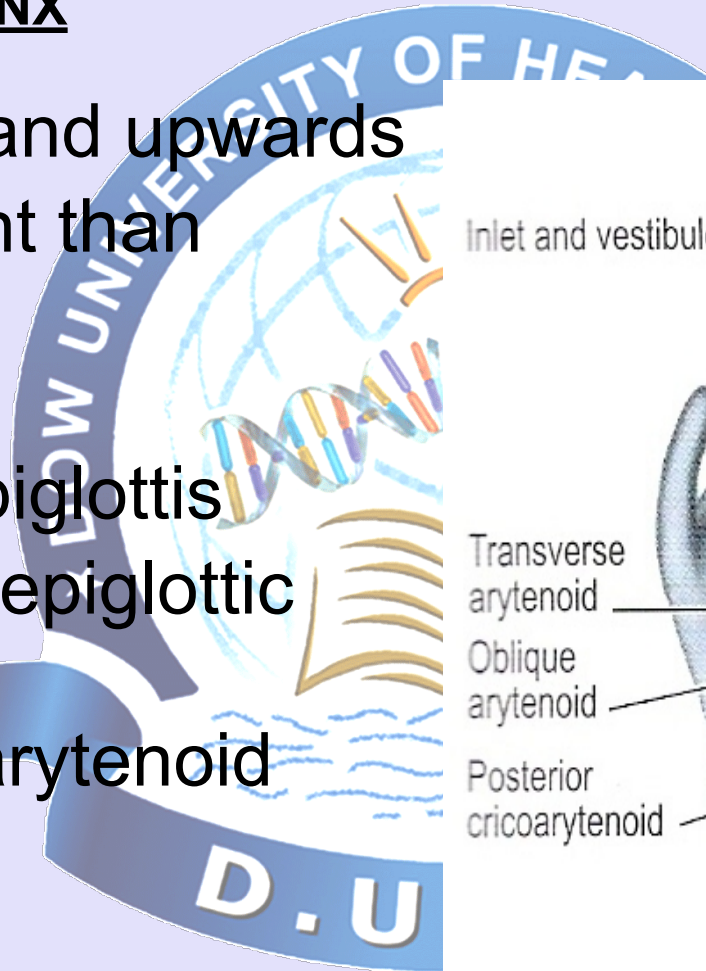
Wider in front than behind

Boundaries

Anteriorly epiglottis

Laterally aryepiglottic folds

Posteriorly arytenoid cartilages

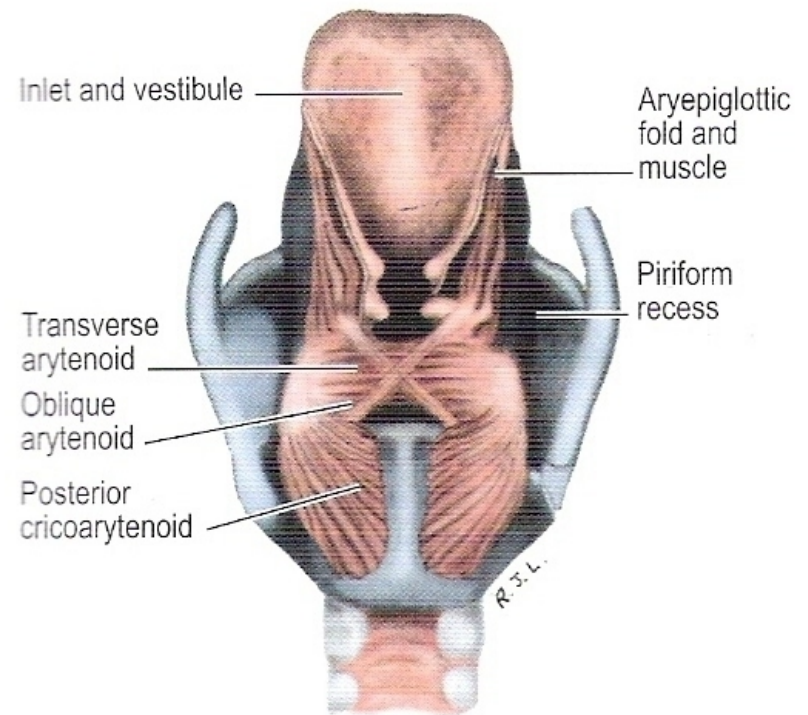


PIRIFORM FOSSA

- Recess on either side of inlet

Boundaries

- Medially
 - Aryepiglottic fold
- Laterally
 - Thyroid cartilage
 - Thyrohyoid membrane

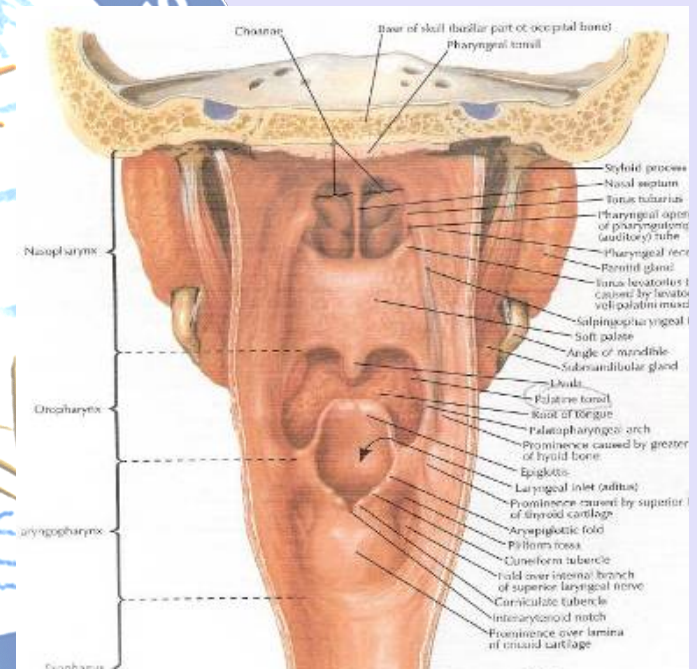


CAVITY OF LARYNX

- Tubular in shape ;lined by mucosa
- Extends from inlet to lower border of cricoid

Boundaries

- **Anterior border** -mucosa covering the superior margin of the epiglottis.
- **Lateral borders** - mucosal folds (aryepiglottic folds), which enclose the superior margins of the quadrangular membranes.
- **Posterior border** in the midline - mucosal fold that forms a depression (interarytenoid notch) between the two corniculate tubercles.

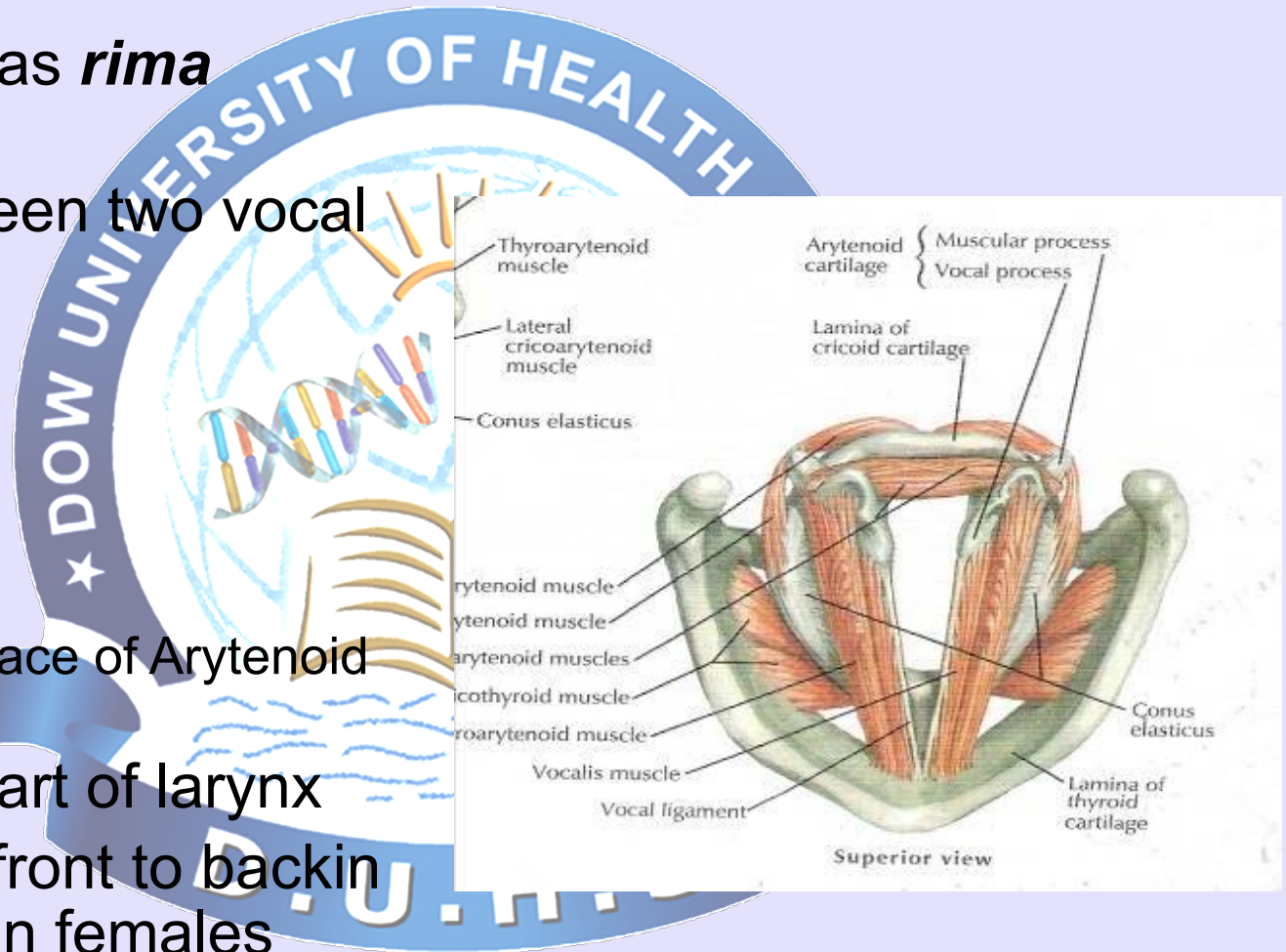


GLOTTIS

- Also known as ***rima glottidis***
- Space between two vocal folds

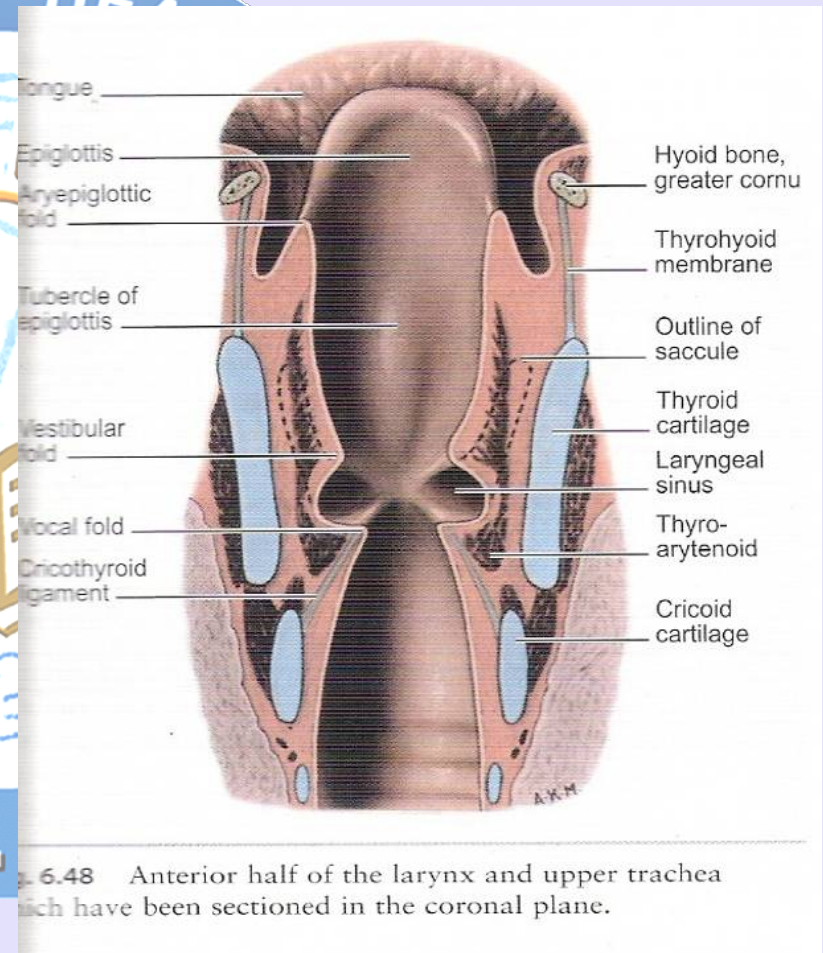
Boundaries

- Anteriorly
 - Vocal folds
- Posteriorly
 - Medial surface of Arytenoid cartilage
- Narrowest part of larynx
- 2.5cm from front to back in males, less in females



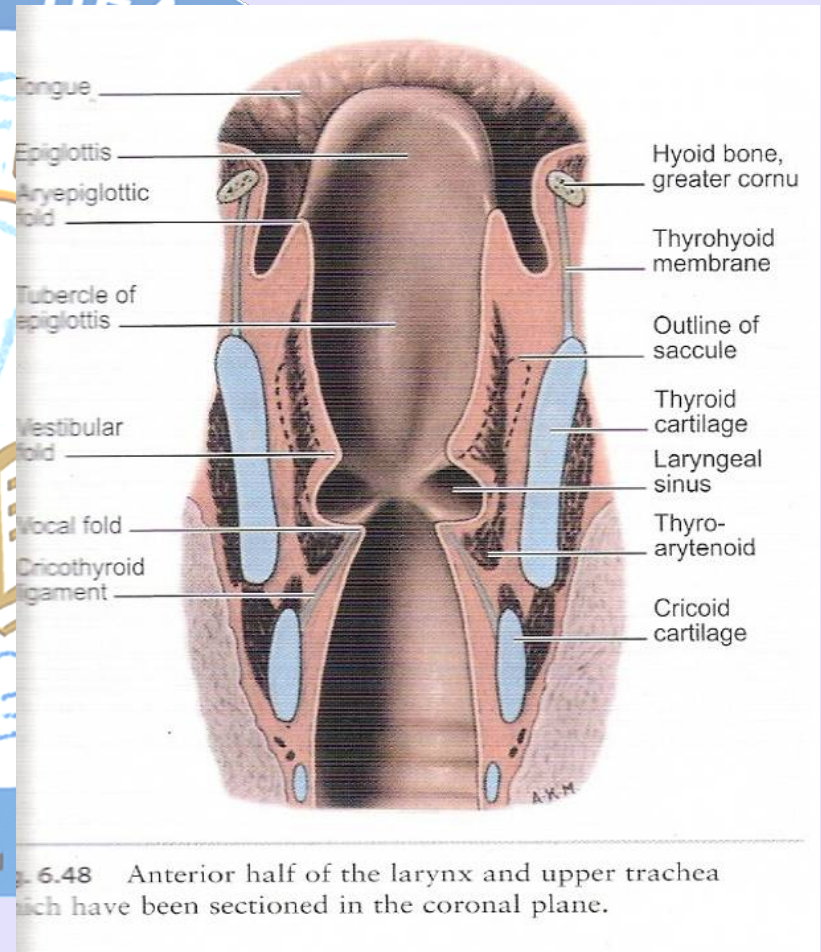
DIVISION OF LARYNGEAL CAVITY

- i. The **vestibule** between the laryngeal inlet and the vestibular folds,.
- ii. The **middle part** of the laryngeal cavity is very thin and is between the vestibular folds above and the vocal folds below.
- iii. The **infraglottic space** between the vocal folds (and the inferior opening of the larynx).



SINUS OF LARYNX

- Part between vocal and vestibular fold is **ventricle of larynx** (sinus).
- The sinus of larynx anterior is prolonged as diverticulum ★ **sacculus of larynx** which contains mucous glands for lubrication of vocal folds.

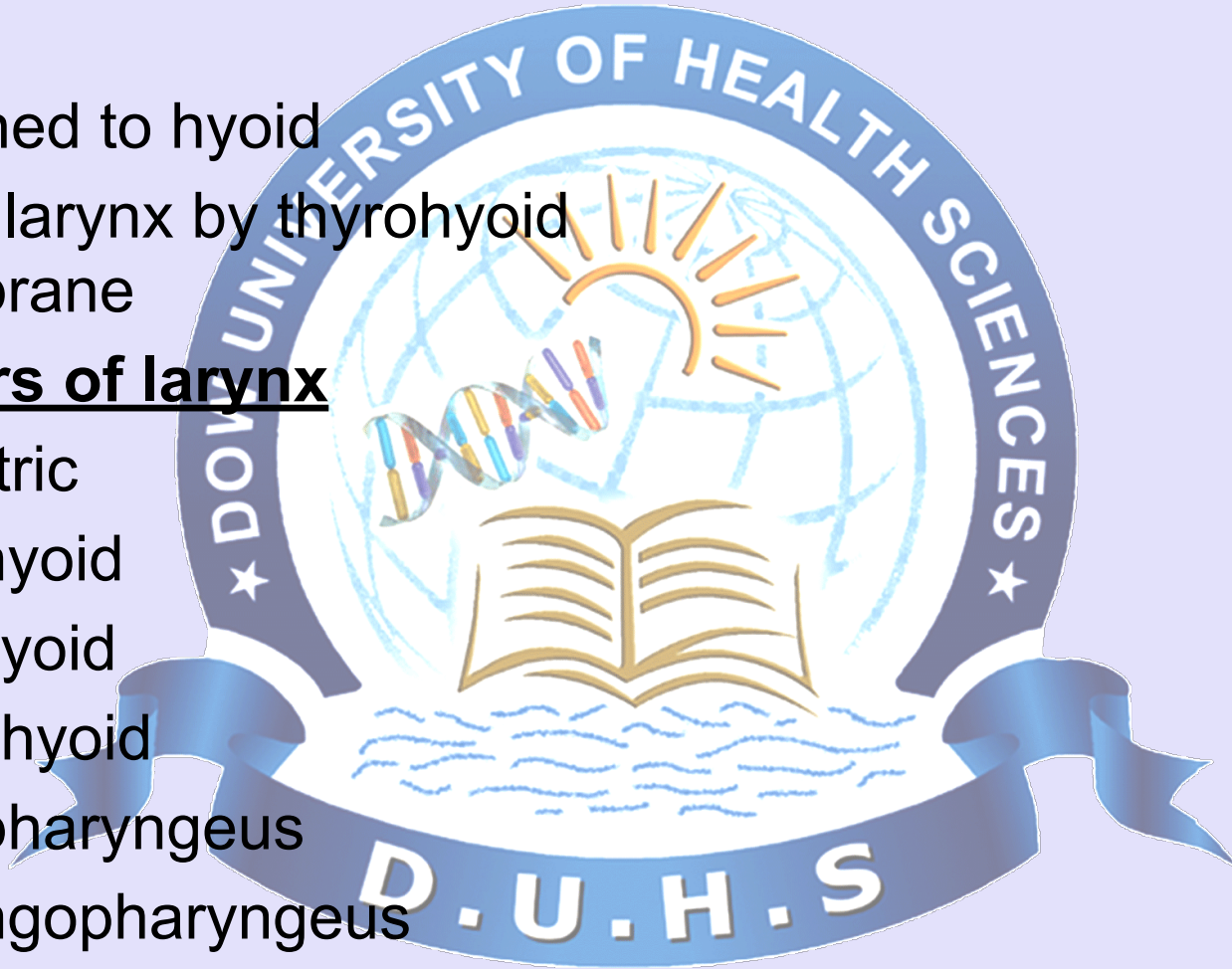


EXTRINSIC MUSCLES

- Attached to hyoid
- Move larynx by thyrohyoid membrane

Elevators of larynx

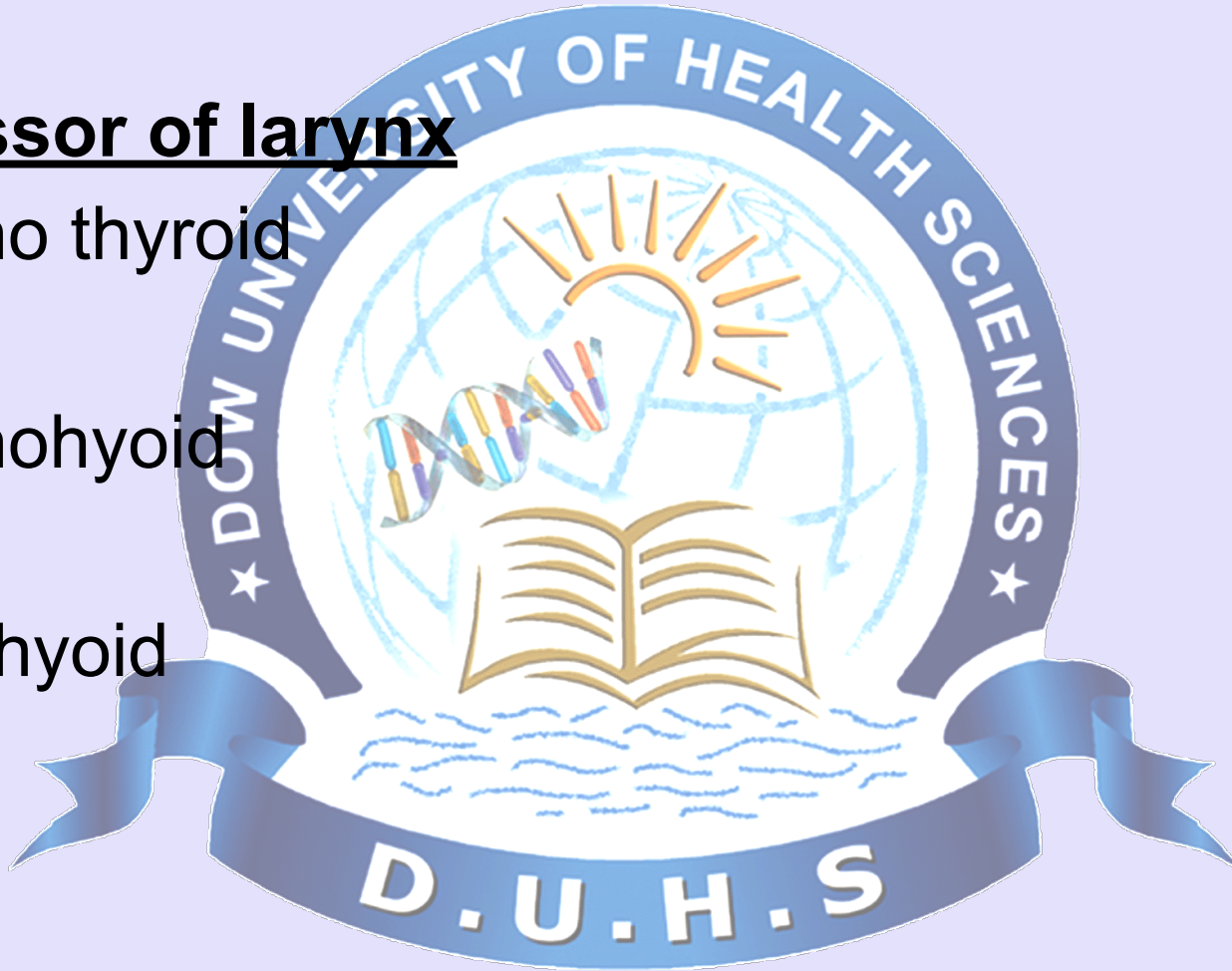
- Digastric
- Stylohyoid
- Mylohyoid
- Geniohyoid
- Stylopharyngeus
- Salpingopharyngeus
- Palatopharyngeus



EXTRINSIC MUSCLES

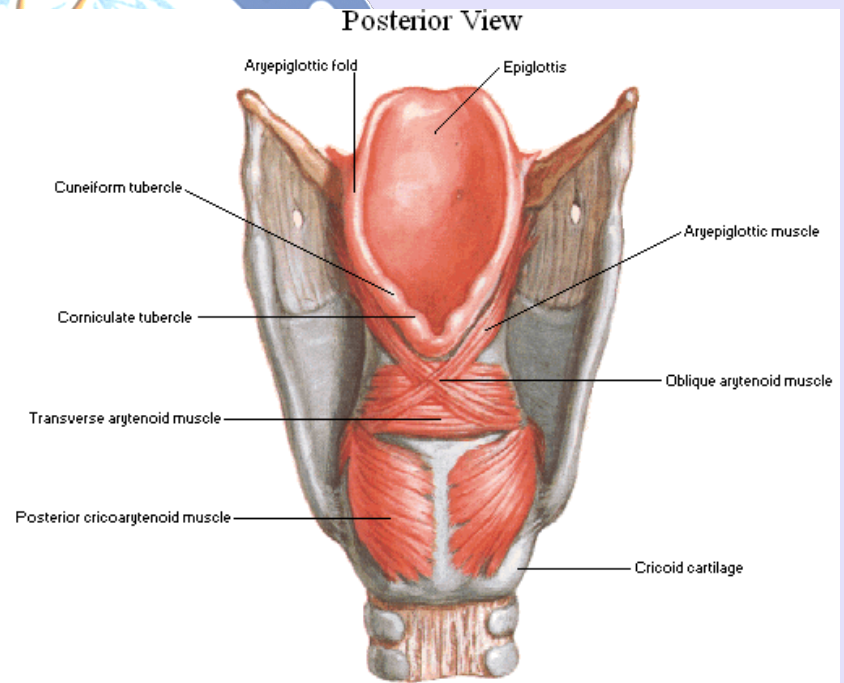
Depressor of larynx

- Sterno thyroid
- Sternohyoid
- Omohyoid



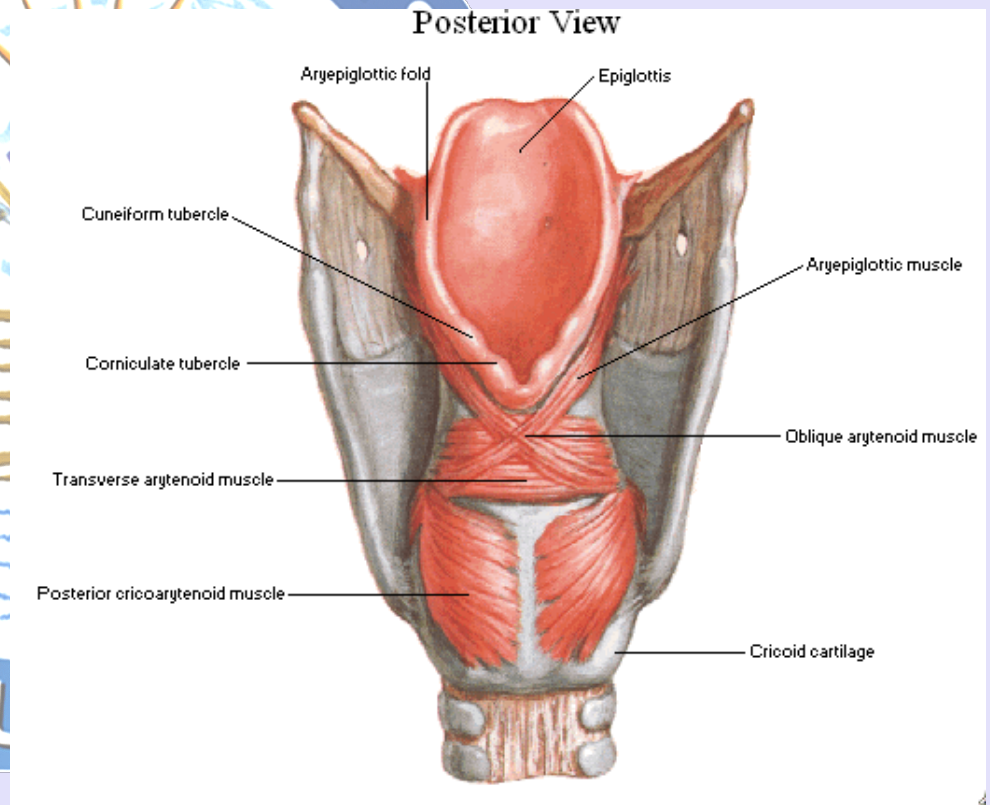
INTRINSIC MUSCLES

- Cricothyroid
- **Lat.** And post. Cricoarytenoid,
- Transverse arytenoid
- Oblique arytenoid
- Thyro arytenoid (vocalis)
- Thyro epiglottic



FUNCTIONS OF INTRINSIC MUSCLES

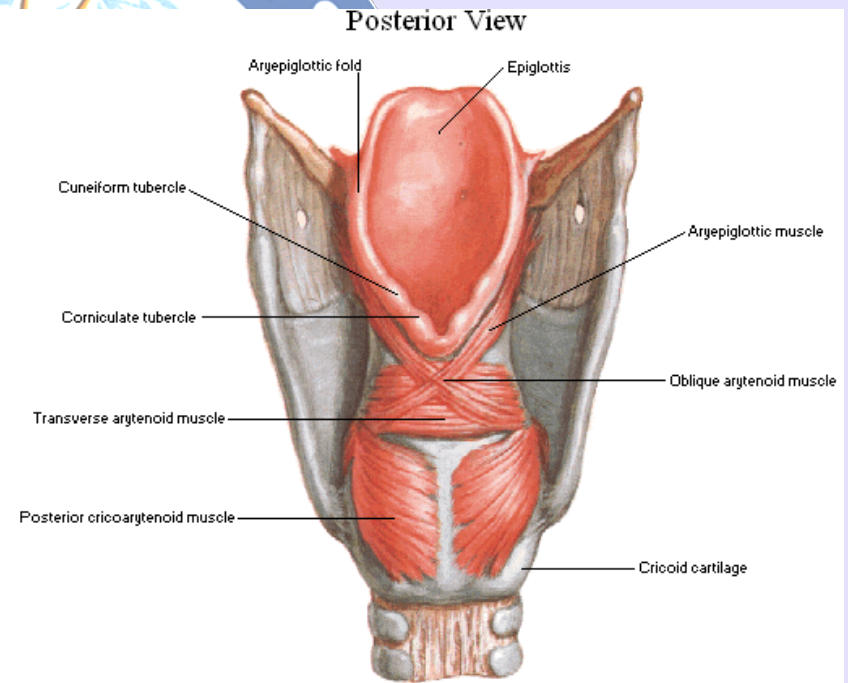
- **MUSCLES CONTROLLING INLET**
- Narrowing-oblique arytenoid
- Widening thyroepiglottic



FUNCTIONS OF INTRINSIC MUSCLES

- MUSCLES MOVING
VOCAL CORDS

- Tensor- cricothyroid
- Relaxor- thyroarytenoid
- Adductor-lateral cricoarytenoid
- Abductor- posterior cricoarytenoid
- Approximate arytenoid-transverse arytenoid



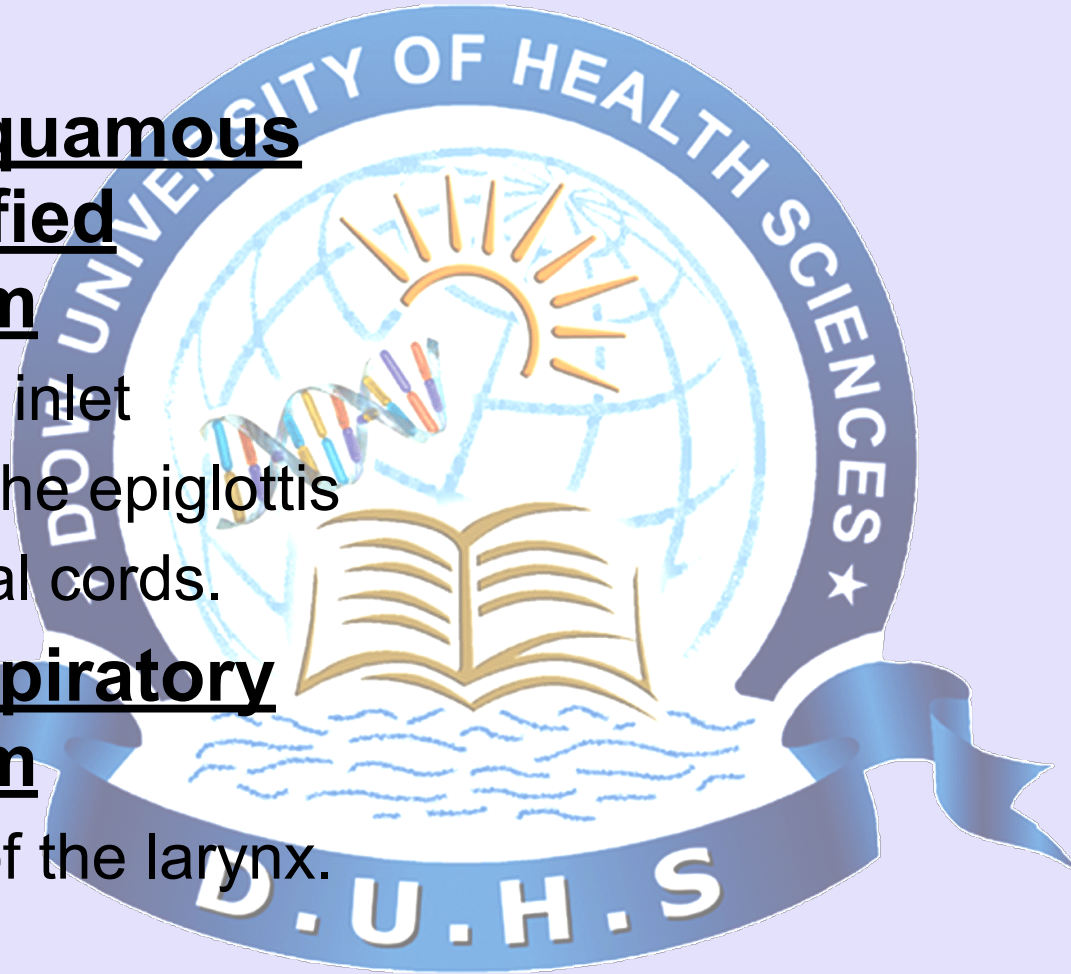
MUCOSA OF LARYNX

stratified squamous noncornified epithelium

- laryngeal inlet
- most of the epiglottis
- true vocal cords.

Typical respiratory epithelium

- the rest of the larynx.



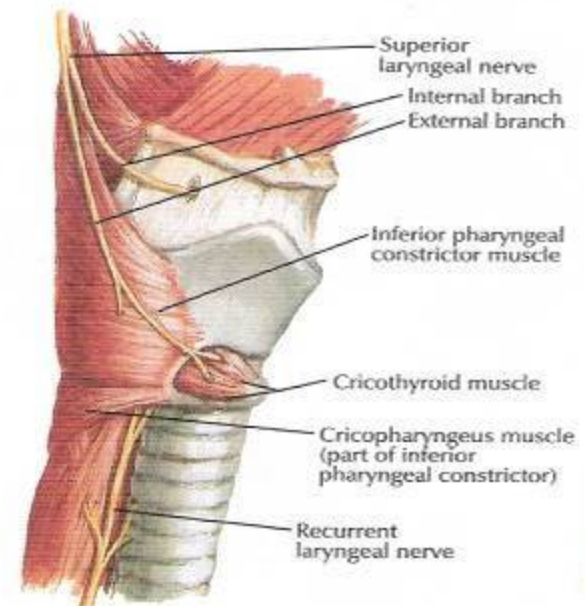
NERVE SUPPLY OF LARYNX

MOTOR

- Recurrent laryngeal nerve except for cricothyroid (external laryngeal nerve)

SENSORY

- Above vocal cords
- Internal laryngeal branch of sup. Laryngeal of vagus
- Below vocal cords
- Recurrent laryngeal



BLOOD SUPPLY OF LARYNX

Upper half

- Superior laryngeal branch of superior thyroid artery

Lower half

- Inferior laryngeal branch of inferior thyroid artery
- Lymphatics- drain into deep cervical group

